$Z\varphi$-classes for extensions of $\mathrm{PerfectGroup}(15000, 3)$

Zphi method from Claude Archer PhD

Solvable quotient groups of orders $2 \le n \le 127$

Here $Z(P) \simeq C_5$, $\mathrm{Out}(P) \simeq C_4$, and $\mathrm{Out}(P) \simeq \mathrm{Aut}(Z(P))$. Thus the four markings of a normal C_5 are equivalent. The columns 1, C_2, and C_4 denote the image of the induced coupling in $\mathrm{Out}(P) \simeq C_4$. The column C_2 records non-surjective couplings. Times are GAP runtimes. Here $|S| = 5n$ and $|E| = 15000n$.

n	$\lvert S\rvert$	$\lvert E\rvert$	1	C_2	C_4	Total	Cumulative	Time (s)	Cumulative time (s)
2	10	30 000	1	1	0	2	2	0.031	0.031
3	15	45 000	1	0	0	1	3	0.002	0.033
4	20	60 000	2	2	1	5	8	0.109	0.142
5	25	75 000	2	0	0	2	10	0.014	0.156
6	30	90 000	2	2	0	4	14	0.282	0.438
7	35	105 000	1	0	0	1	15	0.003	0.441
8	40	120 000	5	7	2	14	29	0.626	1.067
9	45	135 000	2	0	0	2	31	0.012	1.079
10	50	150 000	3	3	0	6	37	0.184	1.263
11	55	165 000	1	0	0	1	38	0.006	1.269
12	60	180 000	5	5	2	12	50	0.902	2.171
13	65	195 000	1	0	0	1	51	0.002	2.173
14	70	210 000	2	2	0	4	55	0.260	2.433
15	75	225 000	2	0	0	2	57	0.042	2.475
16	80	240 000	14	28	9	51	108	6.467	8.942
17	85	255 000	1	0	0	1	109	0.003	8.945
18	90	270 000	5	5	0	10	119	0.765	9.710
19	95	285 000	1	0	0	1	120	0.003	9.713
20	100	300 000	7	8	5	20	140	1.020	10.733
21	105	315 000	2	0	0	2	142	0.201	10.934
22	110	330 000	2	2	0	4	146	0.234	11.168
23	115	345 000	1	0	0	1	147	0.005	11.173
24	120	360 000	15	24	5	44	191	5.783	16.956
25	125	375 000	6	0	0	6	197	0.227	17.183
26	130	390 000	2	2	0	4	201	0.275	17.458
27	135	405 000	5	0	0	5	206	0.085	17.543
28	140	420 000	4	5	2	11	217	1.325	18.868
29	145	435 000	1	0	0	1	218	0.020	18.888
30	150	450 000	6	6	0	12	230	1.440	20.328
31	155	465 000	1	0	0	1	231	0.051	20.379
32	160	480 000	51	144	40	235	466	42.162	62.541
33	165	495 000	1	0	0	1	467	0.026	62.567
34	170	510 000	2	2	0	4	471	0.341	62.908
35	175	525 000	2	0	0	2	473	0.020	62.928
36	180	540 000	14	16	6	36	509	6.139	69.067
37	185	555 000	1	0	0	1	510	0.042	69.109
38	190	570 000	2	2	0	4	514	0.207	69.316
39	195	585 000	2	0	0	2	516	0.082	69.398
40	200	600 000	19	32	12	63	579	4.852	74.250
41	205	615 000	1	0	0	1	580	0.020	74.270
42	210	630 000	6	6	0	12	592	1.515	75.785
43	215	645 000	1	0	0	1	593	0.005	75.790
44	220	660 000	4	5	2	11	604	2.072	77.862
45	225	675 000	4	0	0	4	608	0.049	77.911
46	230	690 000	2	2	0	4	612	0.215	78.126
47	235	705 000	1	0	0	1	613	0.006	78.132
48	240	720 000	52	118	28	198	811	41.843	119.975
49	245	735 000	2	0	0	2	813	0.008	119.983
50	250	750 000	11	10	0	21	834	1.700	121.683
51	255	765 000	1	0	0	1	835	0.027	121.710
52	260	780 000	5	6	4	15	850	2.130	123.840
53	265	795 000	1	0	0	1	851	0.004	123.844
54	270	810 000	15	15	0	30	881	5.456	129.300
55	275	825 000	4	0	0	4	885	0.166	129.466
56	280	840 000	13	22	5	40	925	6.397	135.863
57	285	855 000	2	0	0	2	927	0.088	135.951
58	290	870 000	2	2	0	4	931	0.276	136.227
59	295	885 000	1	0	0	1	932	0.004	136.231
60	300	900 000	18	21	10	49	981	5.520	141.751
61	305	915 000	1	0	0	1	982	0.059	141.810
62	310	930 000	2	2	0	4	986	0.361	142.171
63	315	945 000	4	0	0	4	990	0.328	142.499
64	320	960 000	267	1 120	243	1 630	2 620	1043.647	1186.146
65	325	975 000	2	0	0	2	2 622	0.049	1186.195
66	330	990 000	4	4	0	8	2 630	3.809	1190.004
67	335	1 005 000	1	0	0	1	2 631	0.006	1190.010
68	340	1 020 000	5	6	4	15	2 646	1.510	1191.520
69	345	1 035 000	1	0	0	1	2 647	0.005	1191.525
70	350	1 050 000	6	6	0	12	2 659	1.275	1192.800
71	355	1 065 000	1	0	0	1	2 660	0.028	1192.828
72	360	1 080 000	50	87	19	156	2 816	39.720	1232.548
73	365	1 095 000	1	0	0	1	2 817	0.005	1232.553
74	370	1 110 000	2	2	0	4	2 821	0.335	1232.888
75	375	1 125 000	8	0	0	8	2 829	0.503	1233.391
76	380	1 140 000	4	5	2	11	2 840	1.099	1234.490
77	385	1 155 000	1	0	0	1	2 841	0.023	1234.513
78	390	1 170 000	6	6	0	12	2 853	1.730	1236.243
79	395	1 185 000	1	0	0	1	2 854	0.001	1236.244
80	400	1 200 000	67	154	62	283	3 137	46.760	1283.004
81	405	1 215 000	15	0	0	15	3 152	1.103	1284.107
82	410	1 230 000	2	2	0	4	3 156	2.060	1286.167
83	415	1 245 000	1	0	0	1	3 157	0.003	1286.170

continued on the next page

n	$\|S\|$	$\|E\|$	1	C_2	C_4	Total	Cumulative	Time (s)	Cumulative time (s)
84	420	1 260 000	15	19	6	40	3 197	7.386	1293.556
85	425	1 275 000	2	0	0	2	3 199	0.022	1293.578
86	430	1 290 000	2	2	0	4	3 203	0.364	1293.942
87	435	1 305 000	1	0	0	1	3 204	0.006	1293.948
88	440	1 320 000	12	22	5	39	3 243	7.109	1301.057
89	445	1 335 000	1	0	0	1	3 244	0.006	1301.063
90	450	1 350 000	15	15	0	30	3 274	7.423	1308.486
91	455	1 365 000	1	0	0	1	3 275	0.006	1308.492
92	460	1 380 000	4	5	2	11	3 286	1.096	1309.588
93	465	1 395 000	2	0	0	2	3 288	0.169	1309.757
94	470	1 410 000	2	2	0	4	3 292	0.333	1310.090
95	475	1 425 000	2	0	0	2	3 294	0.018	1310.108
96	480	1 440 000	231	797	152	1 180	4 474	543.805	1853.913
97	485	1 455 000	1	0	0	1	4 475	0.005	1853.918
98	490	1 470 000	5	5	0	10	4 485	1.184	1855.102
99	495	1 485 000	2	0	0	2	4 487	0.044	1855.146
100	500	1 500 000	30	34	22	86	4 573	10.624	1865.770
101	505	1 515 000	1	0	0	1	4 574	0.089	1865.859
102	510	1 530 000	4	4	0	8	4 582	3.083	1868.942
103	515	1 545 000	1	0	0	1	4 583	0.002	1868.944
104	520	1 560 000	14	25	10	49	4 632	7.539	1876.483
105	525	1 575 000	4	0	0	4	4 636	0.390	1876.873
106	530	1 590 000	2	2	0	4	4 640	0.311	1877.184
107	535	1 605 000	1	0	0	1	4 641	0.006	1877.190
108	540	1 620 000	45	54	18	117	4 758	32.182	1909.372
109	545	1 635 000	1	0	0	1	4 759	0.007	1909.379
110	550	1 650 000	10	8	0	18	4 777	4.292	1913.671
111	555	1 665 000	2	0	0	2	4 779	0.100	1913.771
112	560	1 680 000	43	110	26	179	4 958	50.924	1964.695
113	565	1 695 000	1	0	0	1	4 959	0.006	1964.701
114	570	1 710 000	6	6	0	12	4 971	1.858	1966.559
115	575	1 725 000	2	0	0	2	4 973	0.019	1966.578
116	580	1 740 000	5	6	4	15	4 988	1.754	1968.332
117	585	1 755 000	4	0	0	4	4 992	0.395	1968.727
118	590	1 770 000	2	2	0	4	4 996	0.382	1969.109
119	595	1 785 000	1	0	0	1	4 997	0.005	1969.114
120	600	1 800 000	62	116	33	211	5 208	45.377	2014.491
121	605	1 815 000	2	0	0	2	5 210	0.444	2014.935
122	610	1 830 000	2	2	0	4	5 214	1.432	2016.367
123	615	1 845 000	1	0	0	1	5 215	0.722	2017.089
124	620	1 860 000	4	5	2	11	5 226	1.433	2018.522
125	625	1 875 000	22	0	0	22	5 248	2.726	2021.248
126	630	1 890 000	16	16	0	32	5 280	6.465	2027.713
127	635	1 905 000	1	0	0	1	5 281	0.006	2027.719
Total for $2 \leq n \leq 127$			1 382	3 156	743	5 281	5 281	2027.719	2027.719